AI Assisted Coding

Name : M.PRAVALIKA

H.no : 2303A52347

Batch : 34

Course Code : 23CS002PC304

Assignment Number : 8.2

Lab 8: Test-Driven Development with AI – Generating and Working with Test Cases

Aim

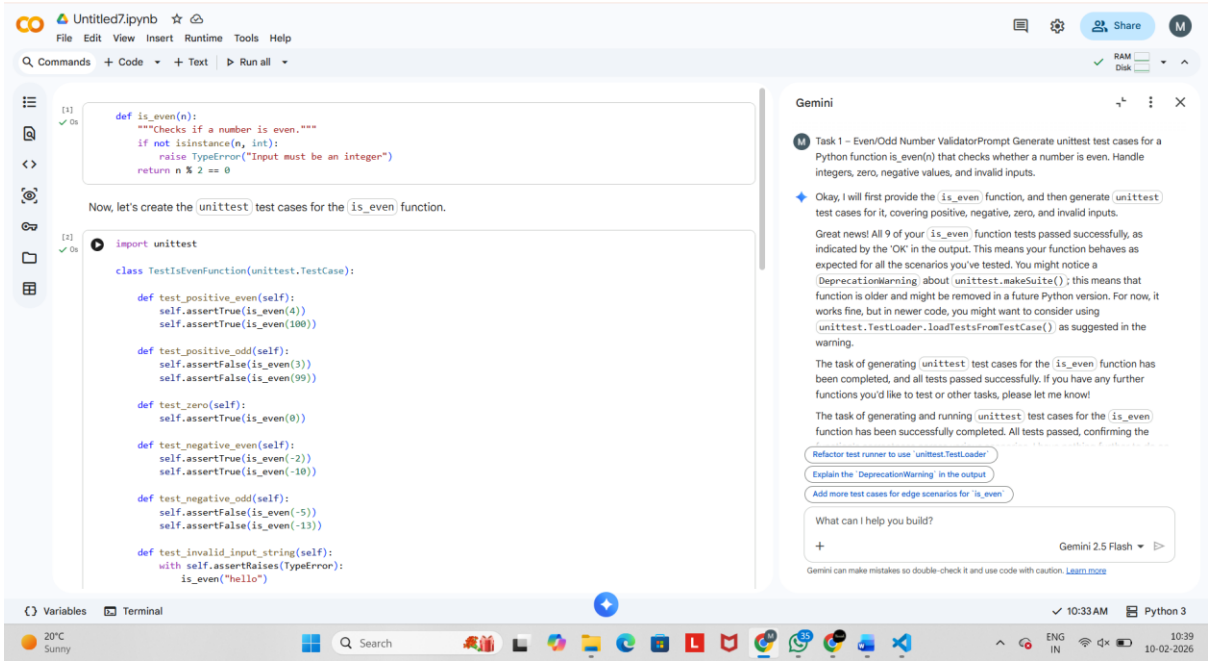
To apply Test-Driven Development (TDD) using AI by generating test cases first and then implementing Python code that satisfies those tests.

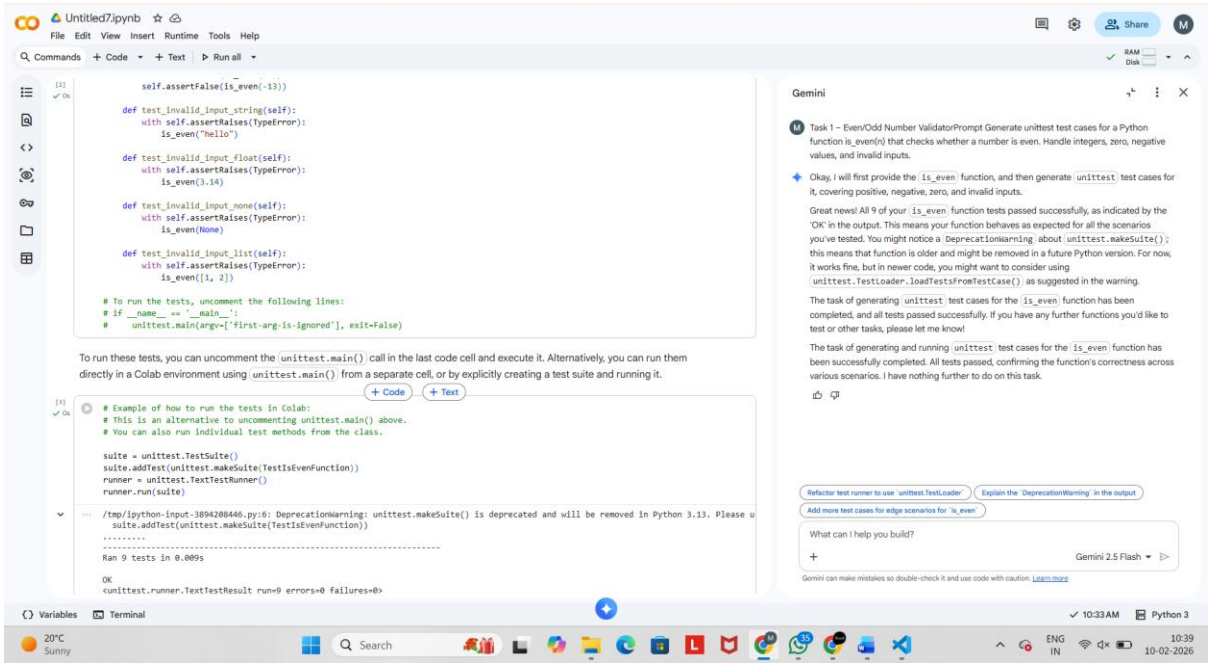
Task Description

Task 1 – Test-Driven Development for Even/Odd Number Validator

Prompt

Generate unittest test cases for a Python function is_even(n) that checks whether a number is even. Handle integers, zero, negative values, and invalid inputs.





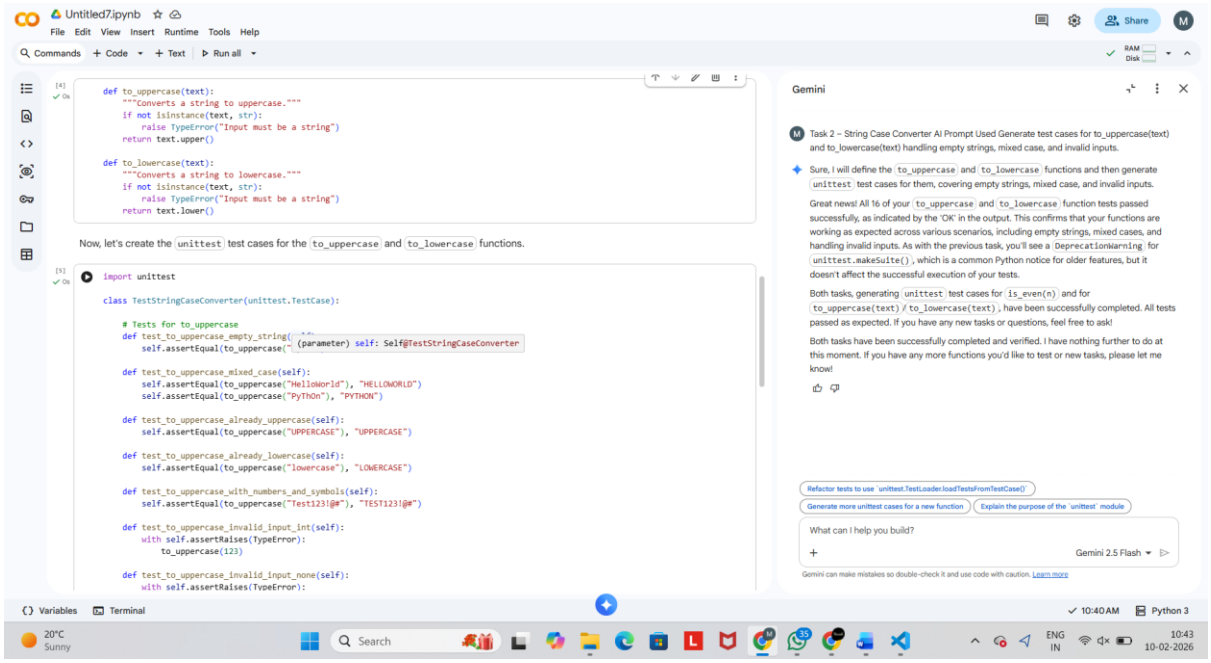
Explanation

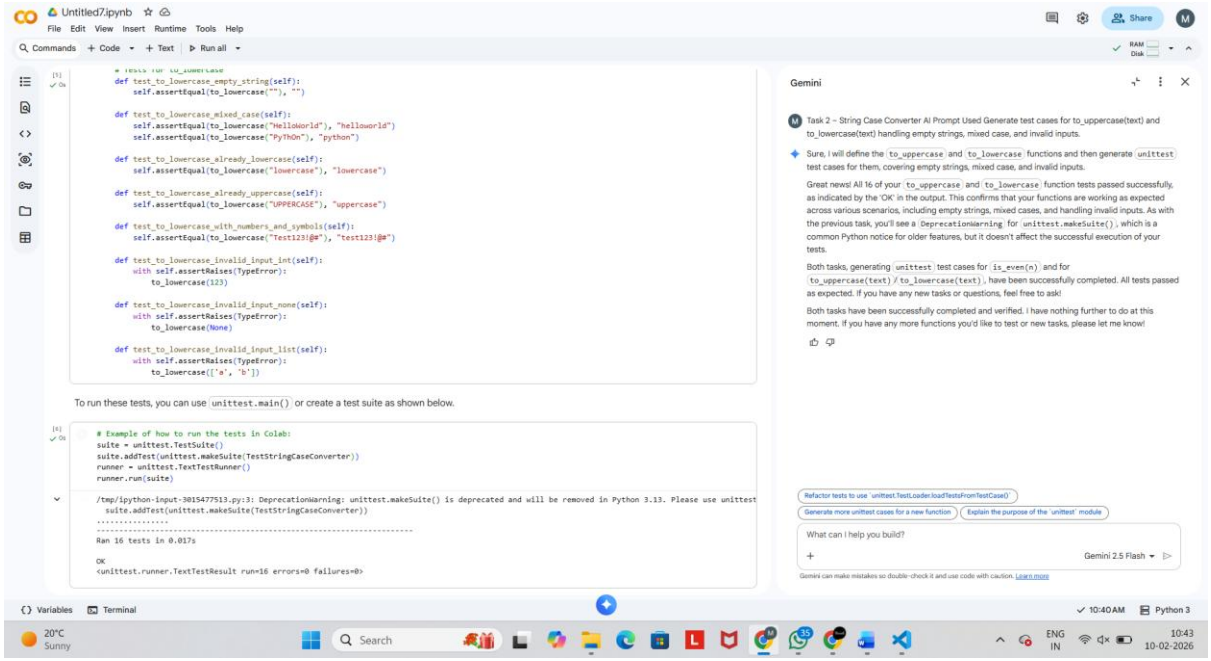
- Validates input type
- Uses modulo operator
- Handles zero, negative, and large integers safely

Task 2 – String Case Converter

Prompt

Generate test cases for to_uppercase(text) and to_lowercase(text) handling empty strings, mixed case, and invalid inputs.





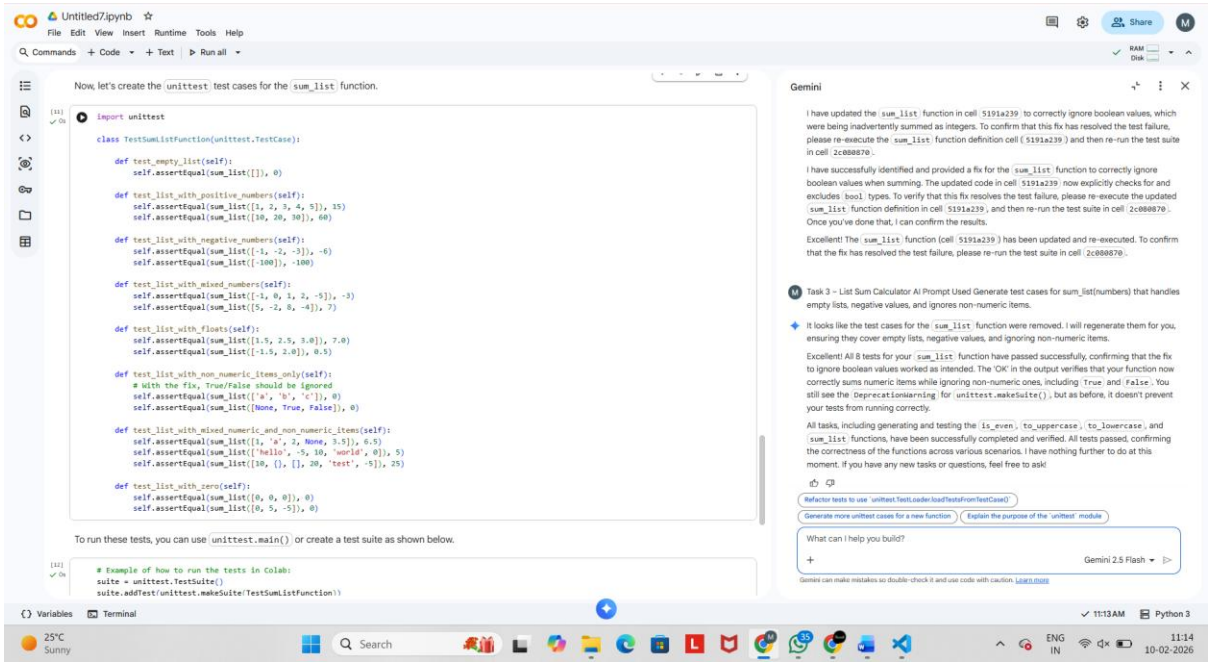
Explanation

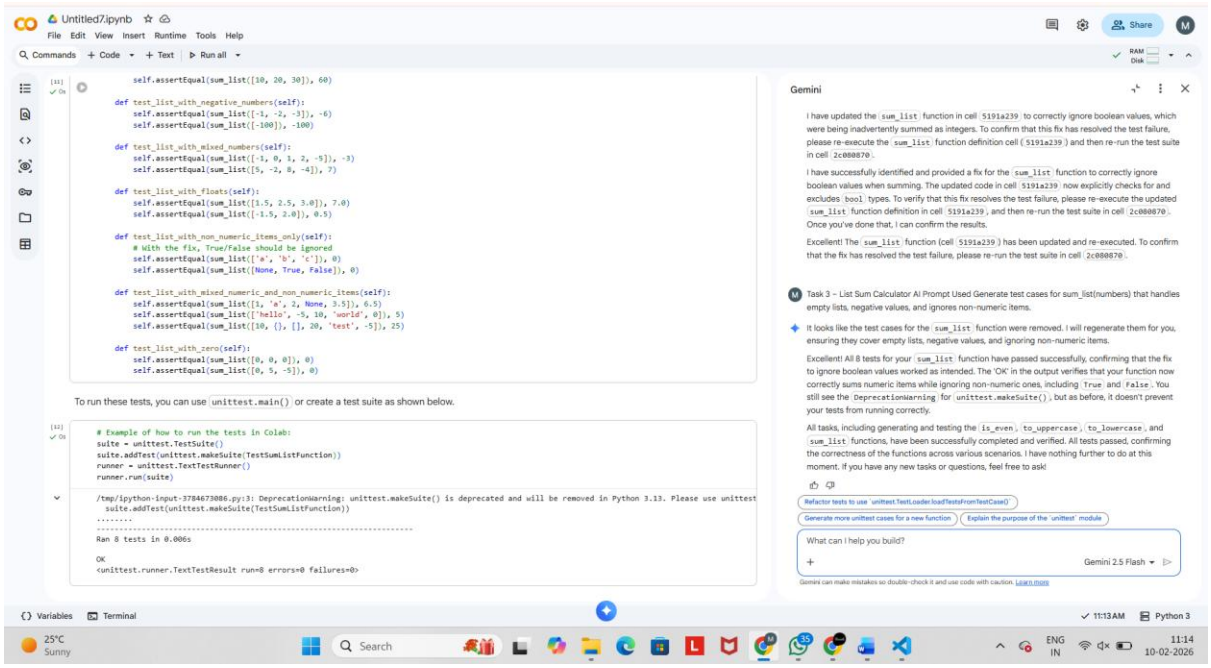
- Safe handling for None
- Uses built-in string methods
- Prevents runtime errors

Task 3 – List Sum Calculator

Prompt

Generate test cases for sum_list(numbers) that handles empty lists, negative values, and ignores non-numeric items.





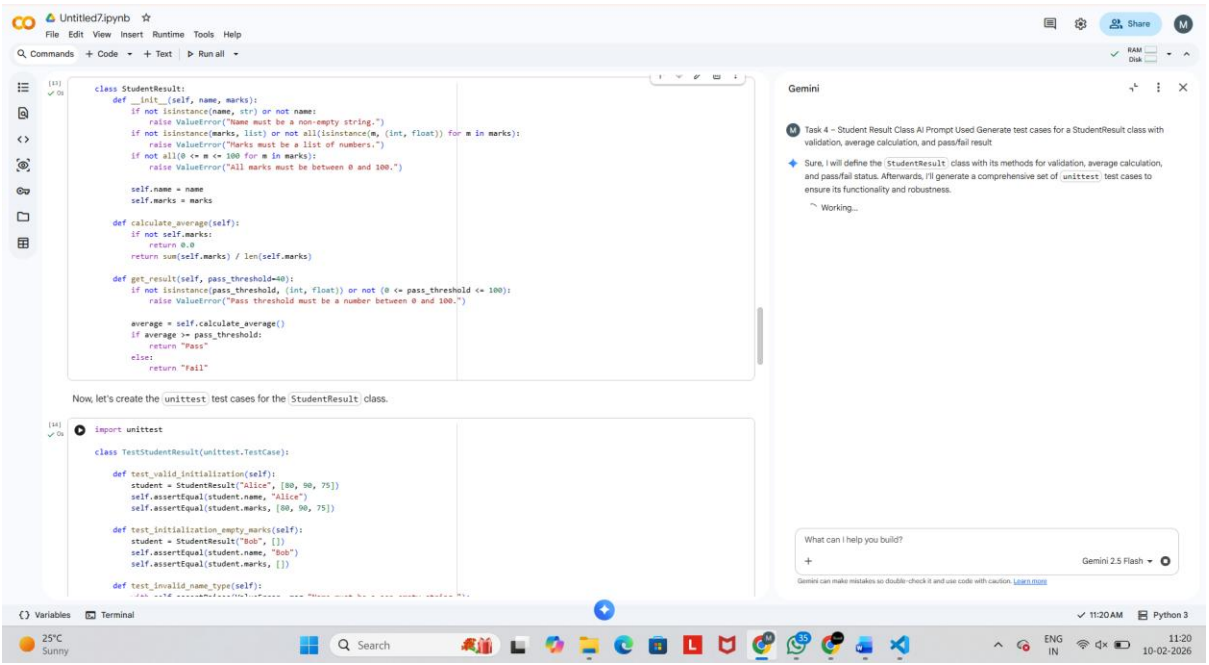
Explanation

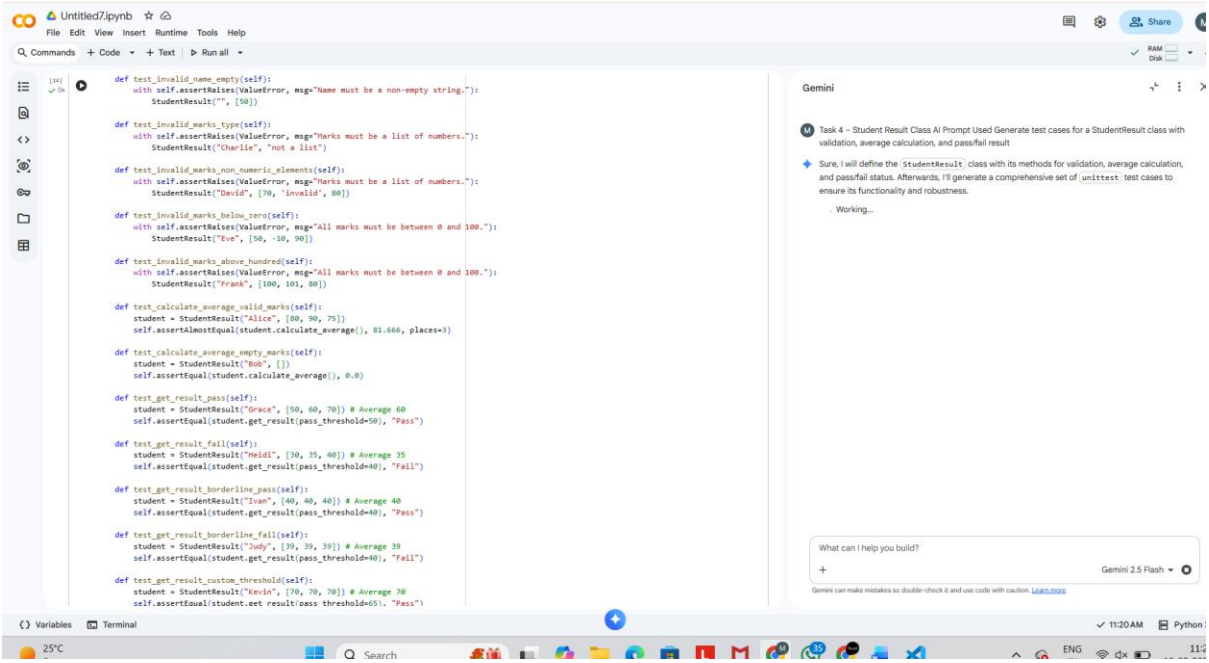
- Iterates through list
- Ignores non-numeric values
- Handles empty lists safely

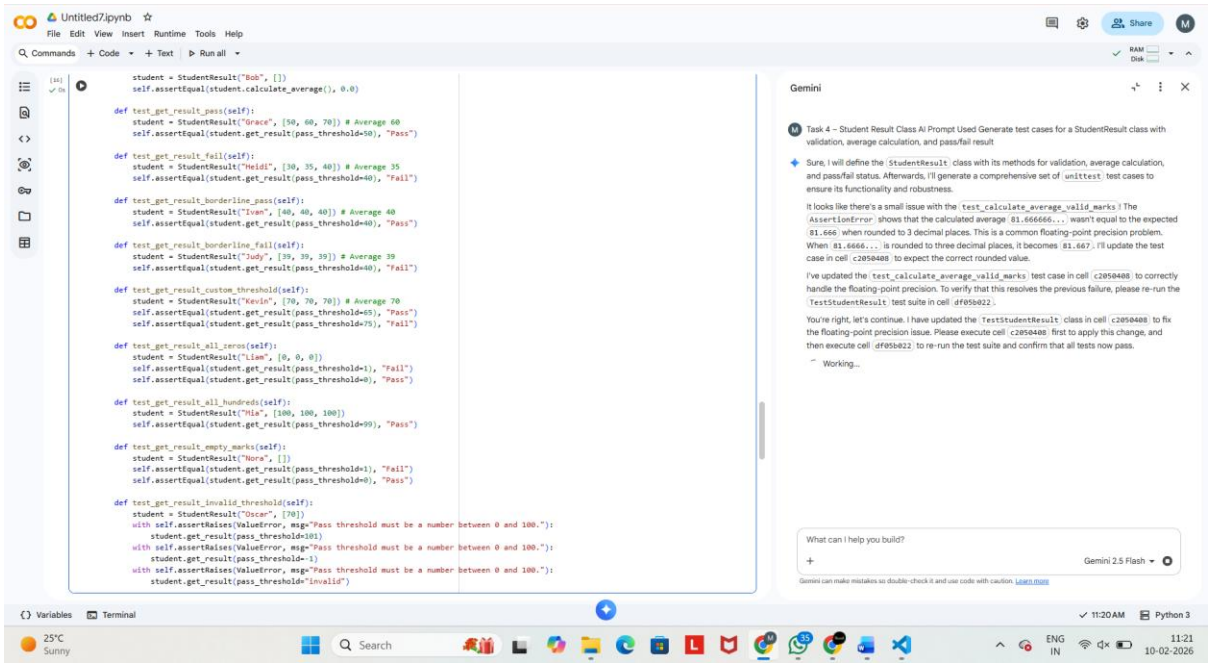
Task 4 – Student Result Class

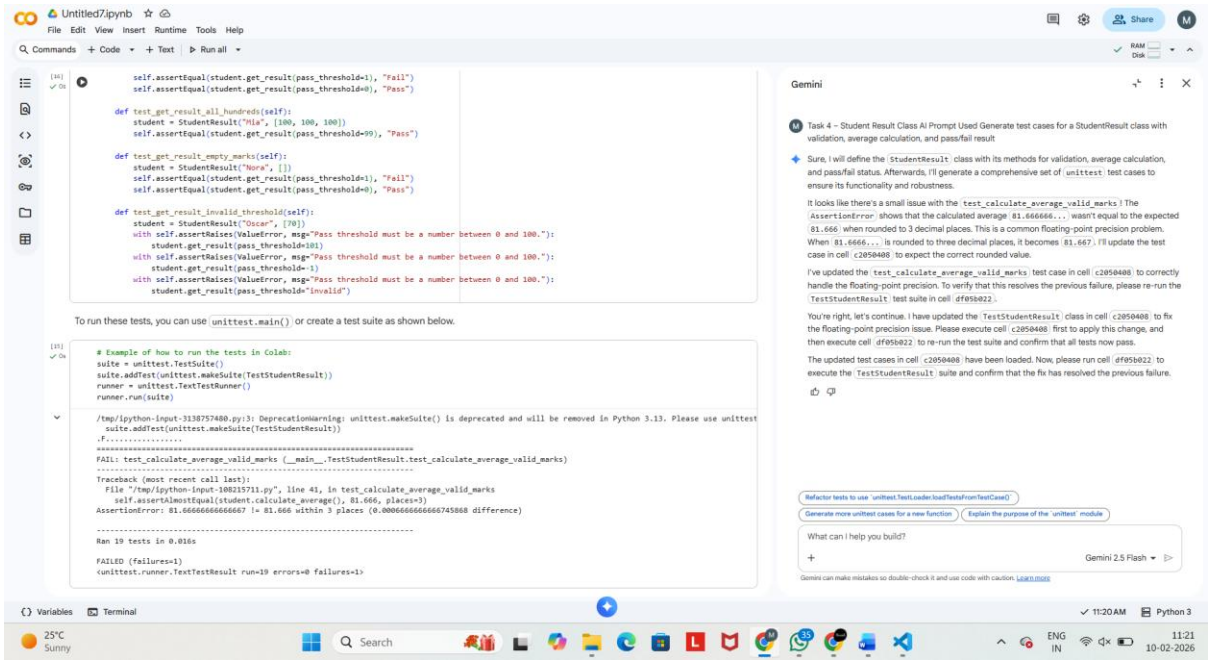
Prompt

Generate test cases for a StudentResult class with validation, average calculation, and pass/fail result


Untitled7.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Share
Gemini
Task 4 – Student Result Class AI Prompt Used Generate test cases for a StudentResult class with validation, average calculation, and pass/fail result
Sure, I will define the StudentResult class with its methods for validation, average calculation, and pass/fail status. Afterwards, I'll generate a comprehensive set of unittest test cases to ensure its functionality and robustness.
Working...
What can I help you build?
Gemini 2.5 Flash
Now, let's create the unittest test cases for the StudentResult class.
Variables Terminal
11:20AM Python 3
25°C Sunny
Search
ENG IN
11:20 10-02-2026


Untitled7.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Share
Gemini
Task 4 – Student Result Class AI Prompt Used Generate test cases for a StudentResult class with validation, average calculation, and pass/fail result
Sure, I will define the StudentResult class with its methods for validation, average calculation, and pass/fail status. Afterwards, I'll generate a comprehensive set of unittest test cases to ensure its functionality and robustness.
Working...
What can I help you build?
Gemini 2.5 Flash
Variables Terminal
11:20AM
25°C
Search
ENG





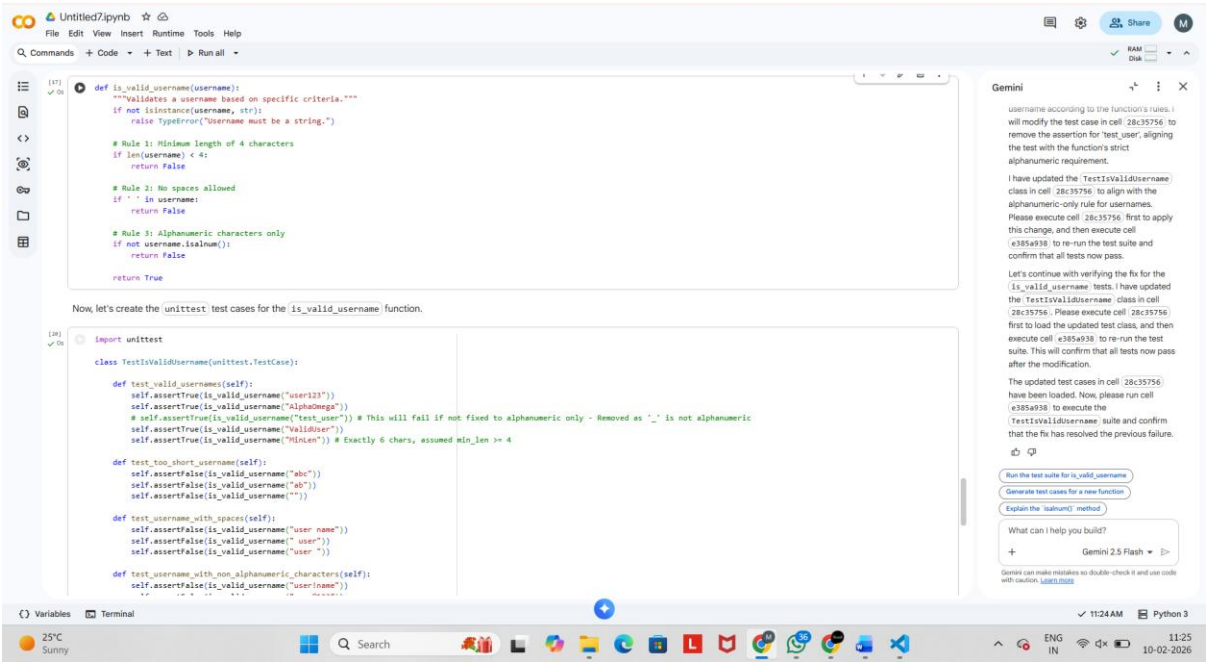
Explanation

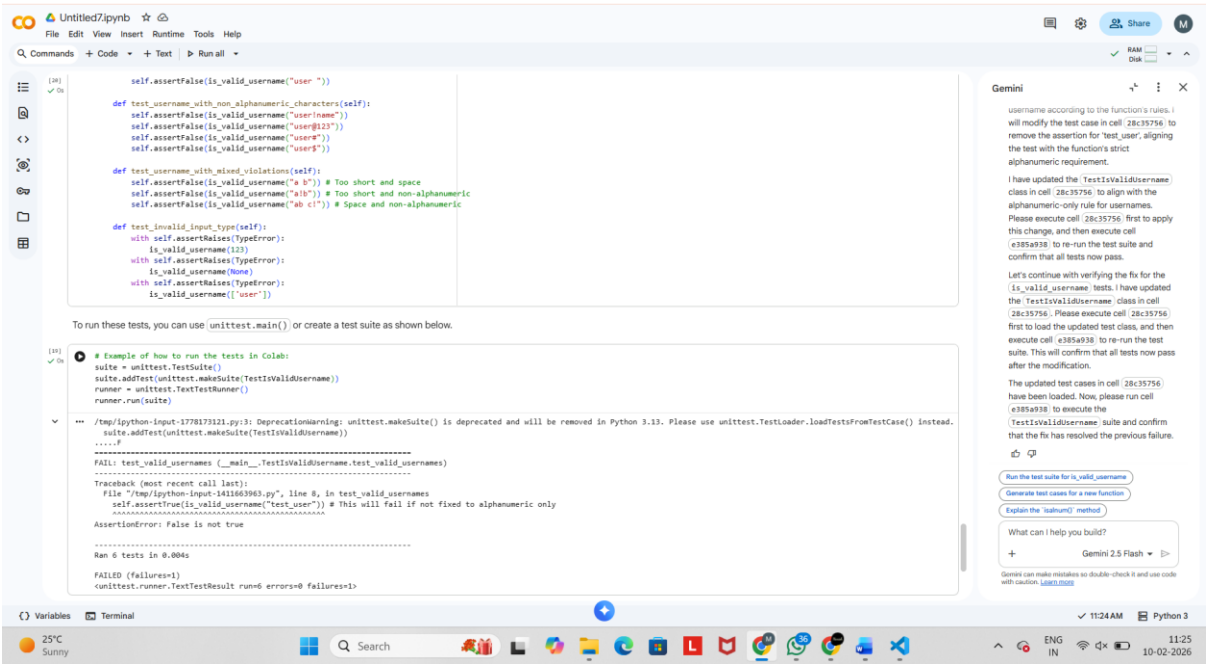
- Input validation enforced
- Average calculated dynamically
- Pass/Fail logic based on requirement

Task 5 – Username Validator

Prompt

Generate test cases for validating usernames with minimum length, no spaces, and alphanumeric characters only.





Explanation

- Length check
- Space validation
- Alphanumeric enforcement

Conclusion

This lab demonstrated:

- Test-Driven Development using AI
- Writing tests before implementation
- Improved reliability and error handling
- Comparison of AI-generated tests with manual logic